



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

FULL SYLLABUS

16-08-2021

JEE(Main + Advanced) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	D	B	A	A	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	D	B	C	D	C	B	C	B	A
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	80.00	0.40	35.00	0.50	80.00	0.60	80.00	4.00	0.25	12.50

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A	A	C	A	A	C	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	A	A	C	D	C	B	C	B	C
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.00	9.00	6.00	7.00	5.00	5.00	1.00	4.00	3.00	3.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	C	A	C	B	C	B	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	B	D	B	D	B	D	D	B	D
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.50	13.00	0.33	0.00	1.00	4.00	0.33	9.00	2.00	72.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (C)**

$$\begin{aligned} \langle \vec{V} \rangle &= \frac{\vec{V}_1 + \vec{V}_2}{2} \\ &= \frac{(10 \cos 37^\circ \hat{i} + 10 \sin 37^\circ \hat{j} + 10 \cos 37^\circ \hat{i})}{2} \\ \Rightarrow \langle \vec{V} \rangle &= 8\hat{i} + 3\hat{j} \end{aligned}$$

2. **Ans (B)**

$$\begin{aligned} F_x - \frac{1}{2}kx^2 - \mu_1 m_1 g x &= 0 \\ Kx &= \mu_2 m_2 g \end{aligned}$$

3. **Ans (C)**

$$\text{For } C, \frac{d^2U}{dx^2} < 0$$

4. **Ans (B)**

$$v = \sqrt{Rg \tan \theta} \Rightarrow \tan \theta = \frac{v^2}{Rg} = \frac{(20)^2}{10 \cdot 10}$$

$$\tan \theta = 4$$

$$\theta = \tan^{-1} 4$$

5. **Ans (D)**

We can write the given force as

$$F = -k \left(y - \frac{C}{k} \right)$$

$$\text{Let us define } s = y - \frac{C}{k} \quad (1)$$

where s is the displacement from mean position because at $y = C/k$, F is zero.Thus, the mean position is at $y = C/k$.

Using Eq. 1, force can be written as

$$F = -ks \quad (2)$$

which is the equation of SHM about $y = C/k$.

Differentiating Eq. 1 with respect to time twice,

we get $\frac{d^2y}{dt^2} = \frac{d^2s}{dt^2}$

Thus, force $F = m(d^2y/dt^2)$ can also be

written as $F = m(d^2s/dt^2)$.

Substituting in Eq. 2 $m \frac{d^2s}{dt^2} = -ks$

Comparing with standard equation we know

that the solution of the above SHM equation is

$$s = y_m \sin(\omega t + \phi)$$

where $\omega = \sqrt{\frac{k}{m}}$

Replacing with value of s from Eq. 1, we get

$$y = y_m \sin(\omega t + \phi) + \frac{C}{k}$$

therefore : this particle is undergoing SHM

about $y = C/k$. Since no information is given

about initial position phase ϕ and amplitude y_m

as unknown.

(b) If in part (a) $F = -10y + 20$, and mass m of

the particle is 2.5 kg and it is released from rest

from $y = +3$, at $t = 0$, write the equation of

SHM.

6. Ans (B)

In same direction and of same frequency

$$x_1 = A_1 \sin \omega t$$

$x_2 = A_2 \sin(\omega t + \theta)$, then resultant displacement

$$x = x_1 + x_2 = A_1 \sin \omega t + A_2 \sin(\omega t + \theta) = A \sin(\omega t + \phi)$$

where $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \theta}$ &

$$\phi = \tan^{-1} \left[\frac{A_2 \sin \theta}{A_1 + A_2 \cos \theta} \right]$$

If $\theta = 0$, both SHMs are in phase and $A = A_1 + A_2$

If $\theta = \pi$, both SHMs are out of phase and

$$A = |A_1 - A_2|$$

The resultant amplitude due to superposition of two or more than two SHMs of this case can also be found by phasor diagram also.

$$x_1 = A_1 \sin \omega_1 t$$

$$x_2 = A_2 \sin \omega_2 t$$

then resultant displacement

$$x = x_1 + x_2 = A_1 \sin \omega_1 t + A_2 \sin \omega_2 t$$

This resultant motion is not SHM.

$$x = A \sin \omega t$$

$$y = B \sin(\omega t + \theta)$$

Case (i) : If $\theta = 0$ or π then $y = \pm (B/A)x$. So path will be straight line & resultant displacement will be $r = (x^2 + y^2)^{1/2} = (A^2 + B^2)^{1/2} \sin \omega t$

which is equation of SHM having amplitude $\sqrt{A^2 + B^2}$

Case (ii) : If $\theta = \frac{\pi}{2}$ then $x = A \sin \omega t$

$$y = B \sin(\omega t + \pi/2) = B \cos \omega t$$

so, resultant will be $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$. i.e.

equation of an ellipse and if $A = B$, then superposition will be an equation of circle.

7. **Ans (A)**

Since soap film on A-side is broken

⇒ Soap film from B-side will pull the thread.

Thus thread will be concave towards A.

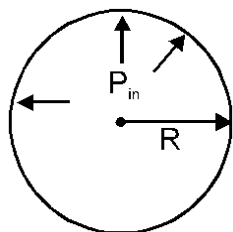
8. **Ans (A)**

we have initially $P_{in} = \frac{4T}{R}$ (1)

Now when extra charge is given,

for equilibrium we have

$$P'_{in} + \frac{\sigma^2}{2\epsilon_0} = \left(\frac{4T}{2R}\right) \dots(1)$$



Now we have $P'_{in} = \frac{(P_{in})v_i}{v_f}$ [isothermal change]

where v_i - initial volume of bubble

v_f - final volume of bubble

$$\text{so, } P'_{in} = \frac{(P_{in})}{8}$$

so, (1) equation will give

$$\frac{4T}{8R} + \frac{\sigma^2}{2\epsilon_0} = \frac{4T}{2R} \Rightarrow \frac{\sigma^2}{2\epsilon_0} = \frac{4T}{2R} \left(1 - \frac{1}{4}\right) = \frac{12T}{8R}$$

$$\text{so, } \sigma^2 = \frac{12T\epsilon_0}{4R} = \left(\frac{3T\epsilon_0}{R}\right) \text{ so, } \sigma = \sqrt{\frac{3T\epsilon_0}{R}}$$

now charge $Q = \sigma \cdot 4\pi (2R)^2 = 4\pi \cdot (4R^2)$

$$\sqrt{\frac{3T\epsilon_0}{R}} = \sqrt{256 \times 3T\epsilon_0 R^2 \pi^2}$$

$$Q = \sqrt{768 \pi^2 R^3 T \epsilon_0}$$

9. **Ans (B)**

$$(x-1)^2 + (y-2)^2 + (z-0)^2 = r^2$$

$$\Rightarrow x^2 + y^2 + z^2 - 3x - 4y + 5 = r^2$$

10. **Ans (B)**

Number for yellow is 4, Number of violet is 7

Brown colour gives multiplier 10^1 ,

Gold gives a tolerance of $\pm 5\%$

So resistance of resistor is $47 \times 10^1 \Omega \pm 5\%$

or $470 \pm 5\% \Omega$.

11. **Ans (B)**

$$B_m = \mu_0 M$$

$$M = \frac{B_m}{\mu_0} = \text{magnetization per unit}$$

volume and volume of one atom

$$= \frac{\text{molecular weight}}{\rho \times N_A}$$

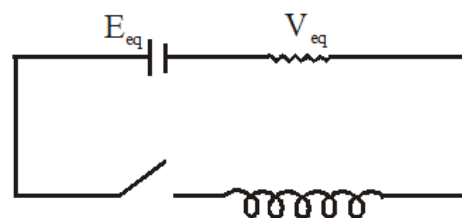
so, magnetization per atom = $M \times \text{volume of}$

$$\text{one atom} = \frac{B_m}{\mu_0} \times \frac{M.W.}{\rho N_A}$$

$$= \frac{1}{4\pi \times 10^{-7}} \times \frac{56 \times 10^{-6}}{7.8 \times 6.023 \times 10^{23}}$$

$$= 9.5 \times 10^{-24} \text{ Am}^2$$

12. **Ans (D)**



$$E_{eq} = \frac{\frac{4}{3} + \frac{-8}{6}}{\frac{1}{3} + \frac{1}{6}} = \frac{\frac{8}{3}}{\frac{1}{2}} = \frac{8}{3} \times \frac{2}{1} = \frac{16}{3} \text{ volt}$$

$$R_{eq} = \frac{3 \times 6}{9} = 2\Omega$$

$$I_{max} = \frac{V}{R} = \frac{16}{3 \times 2} = \frac{8}{3} \text{ A}$$

$$I = \frac{8}{3} (1 - e^{-1000t})$$

13. Ans (B)

$$z = \sqrt{R^2 + x_L^2}$$

$$z = \sqrt{(100)^2 + (100)^2}$$

$$= \sqrt{2}(100)$$

$$\cos \phi = \frac{R}{z} = \frac{100}{100\sqrt{2}} = \frac{1}{\sqrt{2}}$$

14. Ans (C)

$$f = mg$$

$$= 20 \times 10^{-3} \times 10$$

$$= 10^{-1} \times 2$$

$$= 2 \times 10^{-1} \text{ N}$$

$$F_{\text{photon}} = fg$$

$$\frac{P}{C} = 2 \times 10^{-1}$$

$$P = 2 \times 10^{-1} \times 3 \times 10^8$$

$$P = 6 \times 10^7 \text{ W}$$

15. Ans (D)

$$\frac{2E}{C} = mv \quad \dots(i)$$

$$\text{COE } \frac{1}{2}mv^2 = mg\ell(1 - \cos \theta)$$

$$\frac{1}{2} \times \frac{4E^2}{m^2C^2} = g\ell(1 - \cos \theta) = g\ell \frac{\theta^2}{2}$$

$$\therefore \theta = \frac{2E}{mC\sqrt{g\ell}}$$

17. Ans (B)

$$m = \frac{D}{f}$$

18. Ans (C)

$$2\mu t \cos r = \frac{\lambda}{2} \text{ (For } I_{\text{max}} \text{ for first time)}$$

$$\Rightarrow t \propto \lambda$$

$$\Rightarrow \frac{t}{t+0.05} = \frac{4000}{6000} = \frac{2}{3}$$

$$\Rightarrow t = 0.1 \mu\text{m}$$

19. Ans (B)

$$A = A_0 \left(\frac{\sin \beta}{\beta} \right)$$

$$\text{or } \frac{A}{A_0} = \frac{\sin \beta}{\beta}$$

$$\text{Now } \beta = \frac{\pi a \sin \theta}{\lambda} = \frac{\pi}{2}$$

$$\therefore \frac{A}{A_0} = \frac{\sin(\frac{\pi}{2})}{(\frac{\pi}{2})} = \frac{2}{\pi}$$

20. Ans (A)

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1 = \beta = 99$$

$$\beta = \frac{I_c}{I_b} = 99 \Rightarrow I_c = 990 \mu\text{A}$$

$$I_E = I_c + I_b = 1000 \mu\text{A} = 1 \text{ mA}$$

SECTION-II

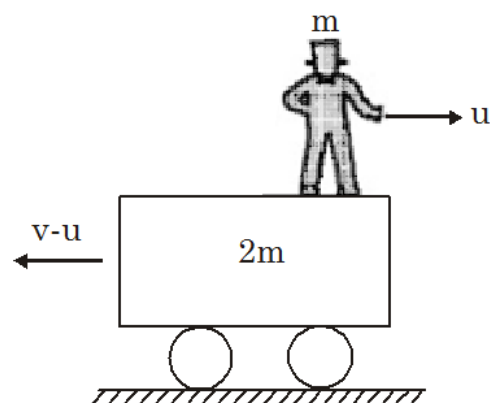
1. Ans (80.00)

Let the velocity of man after jumping be 'u' towards right. Then speed of cart is (v-u) towards left. From conservation of momentum

$$mu = 2m(v - u)$$

$$\therefore u = \frac{2v}{3}$$

hence work done by man = change in K.E. of system



$$= \frac{1}{2}mu^2 + \frac{1}{2}2m(v-u)^2$$

$$= \frac{1}{2}m\left(\frac{2v}{3}\right)^2 + \frac{1}{2}2m\left(\frac{v}{3}\right)^2$$

$$= \frac{mv^2}{3} = \frac{60 \times 2^2}{3} = 80 \text{ J}$$

2. **Ans (0.40)**

$$I = Mv_{cm}$$

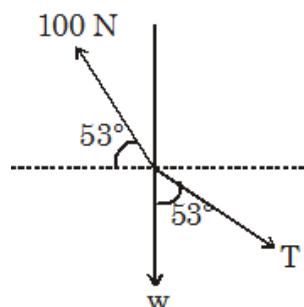
Angular impulse equation about point of contact

$$I(h+r) = \frac{7}{5}Mr^2\omega$$

$$v_{cm} = \omega r$$

$$\frac{h}{r} = \frac{2}{5}$$

3. **Ans (35.00)**



$$T \sin 53^\circ - 100 \cos 53^\circ = 0 \Rightarrow T = 75 \text{ N}$$

$$\text{Also } 100 \sin 53^\circ - T \cos 53^\circ - W = 0$$

$$\Rightarrow W = 35 \text{ N}$$

4. **Ans (0.50)**

$$\text{For equilibrium } T_1 \cdot \frac{\ell}{2} = T_2 \cdot \frac{\ell}{4}$$

$$\frac{T_1}{T_2} = \frac{1}{2}$$

5. **Ans (80.00)**

$$\frac{E_{out}}{E_{in}} = \frac{\frac{kq}{r_{out}^2}}{\frac{kq_{rin}}{R^3}} = \frac{R^3}{r_{in} \times r_{out}^2} = \frac{R^3}{\frac{R}{2} \times \frac{9R^2}{4}}$$

$$\frac{E_{out}}{90} = \frac{8}{9}$$

$$\therefore E_{out} = \frac{8}{9} \times 90 = 80 \text{ N/c}$$

6. **Ans (0.60)**

$$B_{coil} = B_H \tan \theta$$

$$\frac{4\pi \times 10^{-7} \times 20 \times i}{2 \times 20 \times 10^{-2}} = 0.34 \times 10^{-4} (\tan 45^\circ)$$

$$i = \frac{3.4}{2\pi} = 0.6 \text{ A}$$

7. **Ans (80.00)**

Initial energy stored in capacitor $2 \mu\text{F}$

$$U_i = \frac{1}{2} 2(V)^2 = V^2$$

Final voltage after switch 2 is ON

$$V_f = \frac{C_1 V_1}{C_1 + C_2} = \frac{2V}{10} = 0.2V$$

Final energy in both the capacitors

$$U_f = \frac{1}{2} (C_1 + C_2) V_f^2 = \frac{1}{2} 10 \left(\frac{2V}{10} \right)^2 = 0.2V^2$$

So energy dissipated

$$= \frac{V^2 - 0.2V^2}{V^2} \times 100 = 80\%$$

8. **Ans (4.00)**

$$\frac{\Delta d}{d} \times 100 = \frac{0.003}{0.3} \times 100$$

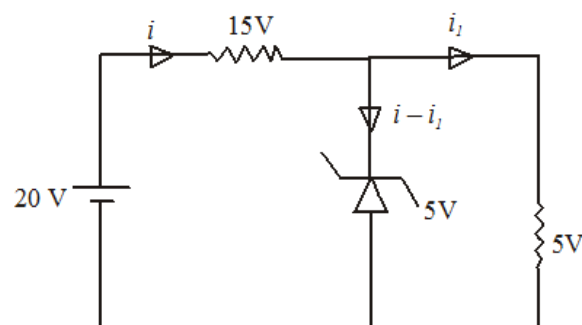
$$+ 2 \times \frac{0.005}{0.5} \times 100 + \frac{0.06}{6} \times 100 = 4\%$$

9. **Ans (0.25)**

$$\text{Power of side band} = \frac{\mu^2}{2} \times P_c$$

$P_c \rightarrow$ power of carrier wave.

10. **Ans (12.50)**

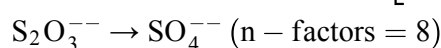
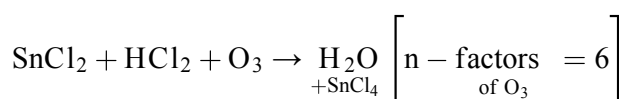
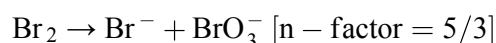
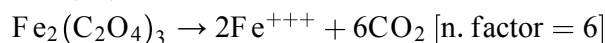


$$i - i_1 = \frac{15}{1000} - \frac{5}{2000} = 12.5 \text{ mA}$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (C)**



2. **Ans (B)**

$$pK_a = \frac{3.0 + 4.4}{2} = 3.7$$

$$\therefore K_{in} = 2 \times 10^{-4}$$

3. **Ans (A)**

Balmer series $n_1 = 1$ α line $n_2 = 3$

β line $n_2 = 4$

$$\frac{1}{\lambda_a} = R \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = \frac{5R}{36}$$

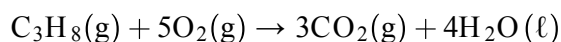
$$\lambda_a = \frac{36}{5R}$$

$$\frac{1}{\lambda_\beta} = R \times 2^2 \left[\frac{1}{2^2} - \frac{1}{4^2} \right] = \frac{3R}{16}$$

$$\lambda_\beta = \frac{16}{3R}$$

$$\frac{\lambda_a}{\lambda_\beta} = \frac{36/5R}{16/3R} = \frac{27}{20}$$

4. **Ans (A)**



$$\Delta n_g = 3 - (1 + 5) = -3$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta H = -2201.1 + \frac{-3 \times 8.314 \times 298}{1000}$$

$$\Delta H = -2201.1 - 7.432$$

$$\Delta H = -2208.53 \text{ KJ}$$

5. **Ans (C)**

CH_3COOH is a weak acid hence it will absorb some amount of energy as condensation enthalpy hence heat energy will decrease.

6. **Ans (A)**

$$A \quad B + C$$

$$100 - x \quad x \quad x$$

$$100 + x = 140$$

$$x = 40$$

$$A = 60 \text{ mm}$$

$$K = \frac{2.303}{30} \log_{10} \frac{100}{60}$$

$$K = \frac{2.303}{30} \times (1 - 0.8)$$

$$K = \frac{2.303 \times 0.2}{30}$$

$$T_{1/2} = \frac{2.303 \times 0.3}{2.303 \times 0.2} \times 30 = 45 \text{ min}$$

7. **Ans (A)**

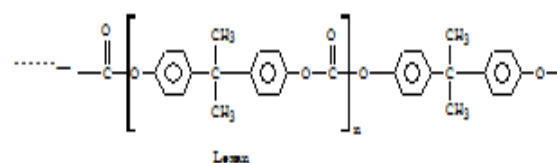
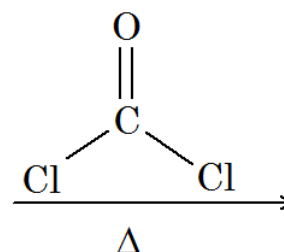
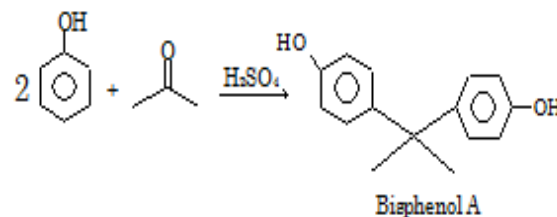
$$E = E^\circ - \frac{0.059}{1} \log_{10} \frac{[Fe^{+3}]}{[Fe^{+2}][Ag^+]}$$

$$0 = (0.8 - 0.77) - \frac{0.059}{1} \log \frac{1}{[Ag^+]}$$

$$\log[Ag^+] = \frac{0.03}{0.059} \approx -\frac{1}{2}$$

$$[Ag^+] = \text{antilog}(-0.5)$$

8. **Ans (C)**



20. **Ans (C)**

Calgon process involves the formation of a soluble complex.

SECTION-II

1. **Ans (3.00)**

meq of $K_2C_2O_4 \cdot 3H_2C_2O_4 \cdot 4H_2O$ = meq of $KMnO_4$

$$0.1 \times V_1 = 2 \times 0.05 \times 5$$

$$V_1 = 5 \text{ mL}$$

meq of $K_2C_2O_4 \cdot 3H_2C_2O_4 \cdot 4H_2O$ = meq of $NaOH$

$$6 \times \frac{0.1}{8} \times 5 = \frac{1}{8} \times 1 \times V_2$$

$$V_2 = 3 \text{ mL}$$

[n-factor of $K_2C_2O_4 \cdot 3H_2C_2O_4 \cdot 4H_2O$ with $KMnO_4$ is 8 and with $NaOH$ is 6]

3. Ans (6.00)

$$\text{for ideal gas } d = \frac{PM}{RT}$$

$$\frac{d}{p} = \frac{M}{RT}$$

$$\text{as per question at } T = 300 \text{ K } \frac{d}{p} = 1$$

$$1 = \frac{M}{0.08 \times 300}$$

$$M = 24 \text{ gm/mole}$$

$$\text{so } M / 4 = 6$$

$\therefore N_2$ remain inert in proces, final volume must be of N_2 only

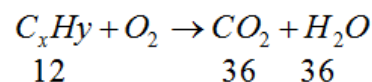
$$V_{N_2} = 360 \text{ mL}$$

$$V_{O_2(\text{initial})} = 450 - 360$$

$$V_{O_2(\text{initial})} = 90 \text{ mL}$$

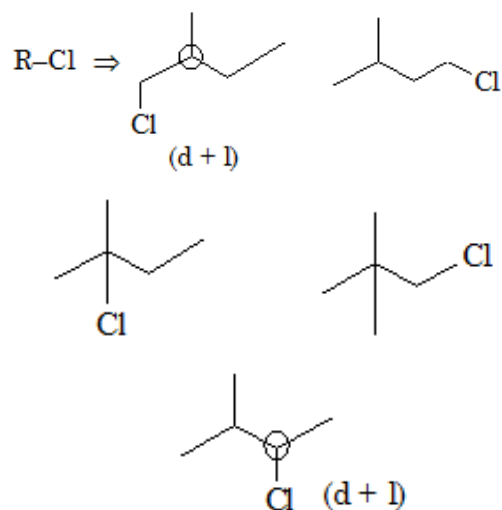
$$V_{O_2 \text{ consumed}} = 54 \text{ mL}$$

$$V_{CO_2} = 36 \text{ mL } V_{H_2O} = 36$$

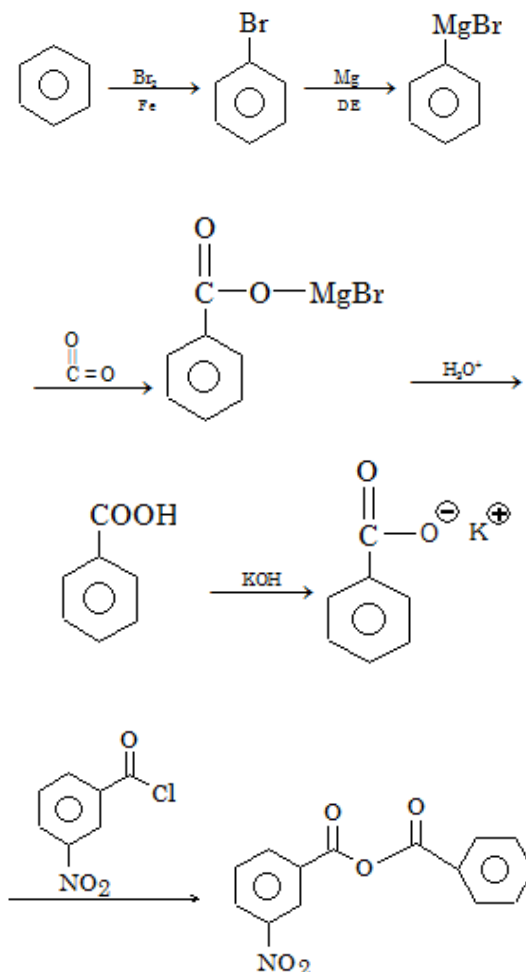


$$\text{formula} = C_3H_6 \text{ so } x + y = 9$$

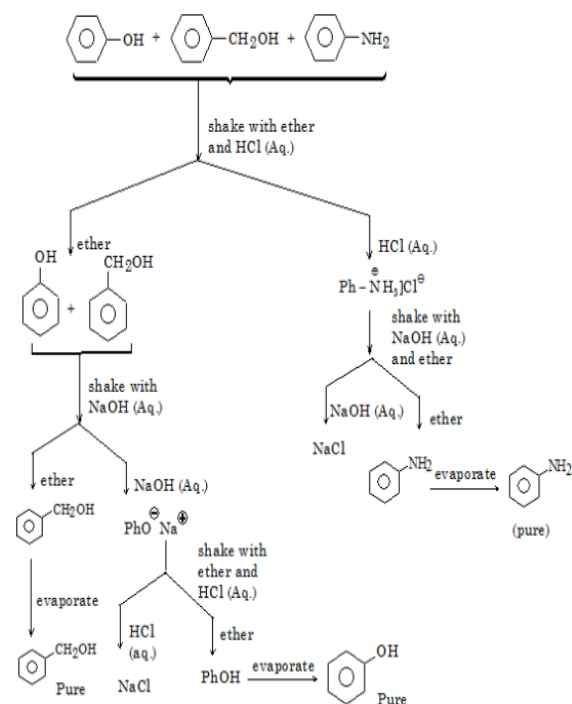
4. Ans (7.00)



5. Ans (5.00)



6. Ans (5.00)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0, \vec{b} \cdot \vec{c} = |\vec{b}| |\vec{c}| \cos \frac{2\pi}{3} = 3 \times 4 \times \frac{-1}{2} = -6$$

$$\text{so, } [\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 4 & 0 & 0 \\ 0 & 9 & -6 \\ 0 & -6 & 16 \end{vmatrix}$$

$$= 4(144 - 36) = 4 \times 108 = 432$$

$$\Rightarrow [\vec{a} \vec{b} \vec{c}] = 2 \times 2 \times 3\sqrt{3} = 12\sqrt{3}$$

2. Ans (C)

p	q	$\sim p \vee q$	$\sim p \wedge \sim q$	$(\sim p \vee q) \wedge (\sim p \wedge \sim q)$
T	T	T	F	F
T	F	F	F	F
F	T	T	F	F
F	F	T	T	T

3. Ans (C)

$$2 \sin x(1 + 2(1 - 2 \sin^2 x))(\cos 3x - \sin 3x) + 1$$

$$= 2(3 \sin x - 4 \sin^3 x)(\cos 3x - \sin 3x) + 1$$

$$= 2 \sin 3x \cos 3x - 2 \sin^2 3x + 1$$

$$= \sin 6x + \cos 6x$$

4. Ans (A)

$$\therefore x = 1 \text{ (domain)}$$

$$\Rightarrow \tan^{-1} \sqrt{\frac{1-y}{1+y}} + \cos^{-1} |y| = \frac{\pi}{4}$$

$$\text{If } y > 0 : \tan^{-1} \sqrt{\frac{1-y}{1+y}} + \cos^{-1} y = \frac{\pi}{4}$$

$$\text{Let } \cos^{-1} y = \theta; \theta \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow \tan^{-1} \left| \tan \frac{\theta}{2} \right| + \theta = \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{\pi}{6} \Rightarrow y = \frac{\sqrt{3}}{2}$$

$$\text{If } y < 0 : \tan^{-1} \sqrt{\frac{1-y}{1+y}} + \pi - \cos^{-1} y = \frac{\pi}{4}$$

$$\text{Let } \cos^{-1} y = \theta; \theta \in \left[\frac{\pi}{2}, \pi\right]$$

$$\Rightarrow \tan^{-1} \left| \tan \frac{\theta}{2} \right| + \pi - \theta = \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{3\pi}{2} \text{ (Not possible)}$$

5. Ans (C)

$$\frac{dy}{dx} = e^{x-y} (e^x - e^y)$$

$$e^y \frac{dy}{dx} + e^x e^y = e^{2x} \dots (1)$$

$$\text{Putting } e^y = v \Rightarrow e^y \frac{dy}{dx} = \frac{dv}{dx}$$

$$\therefore \frac{dv}{dx} + ve^x = e^{2x}$$

$$\text{I. F.} = e^{\int e^x dx} = e^{e^x}$$

$$\text{So, } ve^{e^x} = \int e^{e^x} \cdot e^{2x} dx$$

$$\Rightarrow ve^{e^x} = (e^x - 1) \cdot e^{e^x} + c$$

$$\Rightarrow e^y = e^x - 1 + ce^{-e^x}$$

$$\text{Since } x = 0, y = 1$$

$$\text{So, } c = e^2 \Rightarrow e^y = e^x - 1 + e^{2-e^x}$$

6. **Ans (B)**

$$a^2 + b^2 = 2 + 2 \cos(\theta_1 + \theta_2)$$

7. **Ans (C)**

∴ circumcenter is (0, 0) and centroid is

$$\left(\frac{3 + \sqrt{5}}{3}, \frac{9 + 2\sqrt{5}}{3} \right)$$

i.e. orthocenter is $(3 + \sqrt{5}, 9 + 2\sqrt{5})$

8. **Ans (B)**

$${}^nC_2 = {}^nC_3 \Rightarrow n = 5$$

9. **Ans (A)**

Let $P(at_1^2, 2at_1), Q(at_2^2, 2at_2)$

Slope of $\overline{PQ}$ is $1 \Rightarrow t_1 + t_2 = 2$

Any normal to the parabola is

$$y + xt = 2at + at^3$$

$$\Rightarrow at^3 + t(2a - h) - k = 0 \text{ (locus is (h,k))}$$

$$\text{So, } t_1 + t_2 + t_3 = 0 \Rightarrow t_3 = -2$$

$$\Rightarrow 8a + 2(2a - h) + k = 0$$

10. **Ans (D)**

Given equation $f(x) = x^2 + ax + 1 = 0$



∴ since roots are real, distinct and lying between 0 and 1.

So,

$$\text{i) } a^2 - 4 > 0$$

$$\text{ii) } 0 < -a/2 < 1$$

$$\text{iii) } f(0) > 0 \text{ and } f(1) > 0$$

Hence, $a \in \phi$

11. **Ans (C)**

Equation of director circles of ellipse and hyperbola are respectively.

$$x^2 + y^2 = a^2 + b^2 \text{ and } x^2 + y^2 = a^2 - b^2$$

$$a^2 + b^2 = 4r^2 \quad \dots(1)$$

$$a^2 - b^2 = r^2 \quad \dots(2)$$

$$\text{So } a^2 = \frac{5r^2}{2}, \quad b^2 = \frac{3r^2}{2}$$

Now

$$e_e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{3r^2}{2} \times \frac{2}{5r^2} = 1 - \frac{3}{5} = \frac{2}{5}$$

;

$$\text{Also, } e_h^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{3}{5} = \frac{8}{5}$$

$$\text{So, } 4e_h^2 - e_e^2 = 4 \times \frac{8}{5} - \frac{2}{5} = \frac{30}{5} = 6$$

12. **Ans (B)**

First place can be filled in only 1 way

similarly last place can be filled in 1 way

remaining 4 places can be filled in 4! ways

$$\Rightarrow \text{required permutation} = 1 \times 4! \times 1$$

$$= 24$$

13. **Ans (D)**

$$\frac{3}{1^2 \cdot 2^2} + \frac{7}{1^3 \cdot 2^3} + \frac{5}{2^2 \cdot 3^2} + \frac{19}{2^3 \cdot 3^3} + \frac{7}{3^2 \cdot 4^2} + \frac{37}{3^3 \cdot 4^3} + \dots$$

20 terms

$$\begin{aligned} &= \left(\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} \dots 10 \text{ terms} \right) \\ &+ \left(\frac{7}{1^3 \cdot 2^3} + \frac{19}{2^3 \cdot 3^3} + \frac{37}{3^3 \cdot 4^3} \dots 10 \text{ terms} \right) \\ &= \left(\frac{2^2 - 1^2}{1^2 \cdot 2^2} + \frac{3^2 - 2^2}{2^2 \cdot 3^2} + \frac{4^2 - 3^2}{3^2 \cdot 4^2} \dots 10 \text{ terms} \right) \\ &+ \left(\frac{2^3 - 1^3}{1^3 \cdot 2^3} + \frac{3^3 - 2^3}{2^3 \cdot 3^3} + \frac{4^3 - 3^3}{3^3 \cdot 4^3} \dots 10 \text{ terms} \right) \\ &= \left(\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{2^2} - \frac{1}{3^2} \dots + \frac{1}{10^2} + \frac{1}{11^2} \right) \\ &+ \left(\frac{1}{1^3} - \frac{1}{2^3} + \frac{1}{2^3} - \frac{1}{3^3} \dots + \frac{1}{10^3} - \frac{1}{11^3} \right) \\ &= 1 - \frac{1}{11^2} + 1 - \frac{1}{11^3} = 2 - \frac{1}{121} - \frac{1}{1331} \\ &= \frac{2650}{1331} \end{aligned}$$

14. Ans (B)

$$2 \sum_{1 \leq i < j \leq 100} (\alpha_i \alpha_j)^5$$

$$= (\alpha_1^5 + \alpha_2^5 + \alpha_3^5 + \alpha_4^5 + \dots)^2$$

$$- (\alpha_1^{10} + \alpha_2^{10} + \alpha_3^{10} + \alpha_4^{10} + \dots) = 0 - 0 = 0$$

Since, $(\alpha_1^r + \alpha_2^r + \dots + \alpha_{100}^r)$

$$= \begin{cases} 100, & \text{if } r = 100k \\ 0, & \text{if } r \neq 100k \end{cases}$$

15. Ans (D)

Let $h(x) = f(f(x))$

So, $h(-x) = f(f(-x)) = f(-f(x))$

$$= -f(f(x)) = -h(x)$$

16. Ans (B)

$$f'(x) = 6x^2 - 6(2 + \lambda)x + 12\lambda$$

$\therefore$ For exactly one local max and min

$f'(x) = 0$ will have two real and unequal roots.

$$\Rightarrow D > 0 \Rightarrow (2 + \lambda)^2 - 8\lambda > 0$$

17. Ans (D)

$$f(t) = a^{t \ell_{nb}}$$

$$g(t) = b^{-t \ell_{na}}$$

$$= (b^{\ell_{na}})^{-t} = (a^{\ell_{nb}})^{-t}$$

$$= a^{-t \ell_{nb}}$$

Now $x = a^{t \ell_{nb}}$ and $y = a^{-t \ell_{nb}}$

$$\Rightarrow xy = 1 \text{ or } f(t).g(t) = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1}{x^2}$$

18. Ans (D)

$$f(-x) = \begin{vmatrix} x & \cos x & e^{x^6} \\ \sin^5 x & x^4 & \sec x \\ \tan^3 x & 10 & 20 \end{vmatrix}$$

$$= -f(x)$$

$\Rightarrow f(x)$ is an odd function.

19. Ans (B)

This can happen, if three lines are real and distinct and acute angle between any pair is $\frac{\pi}{3}$.

$$\Rightarrow f(m) = bm^3 + dm^2 + cm + a \text{ has 3 distinct}$$

real roots and

$$\frac{m_1 - m_2}{1 + m_1 m_2} = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{m_3 - m_1}{1 + m_3 m_2} = \pm \sqrt{3}$$

$$\Rightarrow 3 + m_1 m_2 + m_2 m_3 + m_3 m_1 = 0$$

$$\Rightarrow 3b + c = 0 \text{ and } d^2 > 3bc$$

(as $f'(m) = 0$ has two real and distinct roots)

20. Ans (D)

$$\text{Let } N = M - \frac{M^2}{3} + \frac{M^3}{9} - \frac{M^4}{27} + \dots \infty$$

$$\therefore -\frac{MN}{3} = 0 - \frac{M^2}{3} + \frac{M^3}{9} - \dots \infty$$

$$\Rightarrow \left(1 + \frac{M}{3}\right)N = M$$

$$\Rightarrow N = \begin{bmatrix} 3 & -9 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ -1 & 4 \end{bmatrix}^{-1} =$$

$$\frac{1}{13} \begin{bmatrix} 3 & -27 \\ -9 & 3 \end{bmatrix} = \frac{3}{13} \begin{bmatrix} 1 & -9 \\ -3 & 1 \end{bmatrix}$$

$$\therefore \alpha = -9, \beta = -3$$

$$\Rightarrow \left| \frac{\alpha}{\beta} \right| = \left| \frac{-9}{-3} \right| = 3$$

SECTION-II

1. Ans (1.50)

$$S.D = \sqrt{\frac{\sum (x_j - 8)^2}{18} - \left(\frac{\sum (x_j - 8)}{18}\right)^2}$$

$$= \sqrt{\frac{45}{18} - \frac{1}{4}} = \sqrt{\frac{90-9}{36}} = \sqrt{\frac{81}{36}} = \frac{9}{6} = \frac{3}{2}$$

2. Ans (13.00)

Any point on the line is
(3λ + 2, 4λ - 1, 12λ + 2)

Given, x - y + z = 5

$$\Rightarrow 3\lambda + 2 - 4\lambda + 1 + 12\lambda + 2 = 5$$

$$\Rightarrow 11\lambda = 0 \Rightarrow \lambda = 0$$

So, P (2, -1, 2) and Q (-1, -5, -10)

Hence, Distance

3. Ans (0.33)

$$x = \frac{1}{2} \left(\sqrt{5} + \frac{1}{\sqrt{5}} \right) \Rightarrow x = \frac{3}{\sqrt{5}}$$

Now,

$$\log_8 \left(\frac{\sqrt{x^2 - 1}}{x - \sqrt{x^2 - 1}} \right) = \log_8 \left(\frac{\sqrt{\frac{9}{5} - 1}}{\frac{3}{\sqrt{5}} - \sqrt{\frac{9}{5} - 1}} \right)$$

$$= \log_8 \left(\frac{2/\sqrt{5}}{1/\sqrt{5}} \right) = \log_8 2 = \frac{1}{3}$$

4. Ans (0.00)

$$\text{Given } \frac{f(x) \cdot f'(x)}{\sqrt{1 - (f(x))^4}} \geq x$$

$$\Rightarrow \frac{2f(x) \cdot f'(x)}{\sqrt{1 - (f(x))^4}} \geq 2x$$

Now if $g(x) = \sin^{-1} \{ (f(x))^2 \} - x^2$ then $g'(x) \geq 0$

$\Rightarrow g(x)$ is a non-decreasing function

$$\Rightarrow \lim_{x \rightarrow \alpha^+} g(x) \leq \lim_{x \rightarrow \beta^-} g(x)$$

$$\Rightarrow \lim_{x \rightarrow \alpha^+} \left(\{ \sin^{-1}(f(x))^2 \} - x^2 \right) \leq \lim_{x \rightarrow \beta^-} \left(\{ \sin^{-1}(f(x))^2 \} - x^2 \right)$$

$$\Rightarrow \frac{\pi}{2} - \alpha^2 \leq \frac{\pi}{6} - \beta^2$$

$$\Rightarrow \alpha^2 - \beta^2 \geq \frac{\pi}{3} \Rightarrow [\alpha^2 - \beta^2] \geq 1$$

5. Ans (1.00)

$$\begin{vmatrix} x^2 & x & 1 \\ x & 1 & x^2 \\ 1 & x^2 & x \end{vmatrix} = 0$$

$$\Rightarrow (x^2 + x + 1) \begin{vmatrix} 1 & 1 & 1 \\ x & 1 & x^2 \\ 1 & x^2 & x \end{vmatrix} = 0 \quad (R_1 \rightarrow R_1 + R_2 + R_3)$$

$$\Rightarrow (x^2 + x + 1) \begin{vmatrix} 0 & 0 & 1 \\ x-1 & 1-x^2 & x^2 \\ -(x^2-1) & -(x-x^2) & x \end{vmatrix} = 0$$

$$\begin{pmatrix} c_1 \rightarrow c_1 - c_2 \\ c_2 \rightarrow c_2 - c_3 \end{pmatrix}$$

$$\Rightarrow (x^2 + x + 1)(x-1)(1-x) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1+x & x^2 \\ -(x+1) & -x & x \end{vmatrix} = 0$$

$$\Rightarrow (x^2 + x + 1)(x-1)^2 \{ x - (x+1)^2 \} = 0$$

$$\Rightarrow (x^2 + x + 1)^2 (x-1)^2 = 0$$

6. Ans (4.00)

Equation of tangent is $y = 2x \pm \sqrt{4a^2 + b^2}$

It is normal to the circle $x^2 + y^2 + 4x + 1 = 0$

Hence, this tangent passes through (-2, 0)

$$\Rightarrow 0 = -4 \pm \sqrt{4a^2 + b^2} \Rightarrow 4a^2 + b^2 = 16$$

Using A.M. $\geq$ G.M, we get

$$\frac{4a^2 + b^2}{2} \geq \sqrt{4a^2 b^2} \Rightarrow ab \leq 4$$

7. Ans (0.33)

There are 45 two digit even numbers. So,

reduced sample space is S and $n(S) = 45$

Now, if an even number is divisible by three then, it must be divisible by six.

So, $E = \{12, 18, \dots, 96\} \Rightarrow n(E) = 15$

$$\text{Hence, } P(E) = \frac{15}{45} = \frac{1}{3}$$

8. Ans (9.00)

$$\begin{aligned}
 P &= {}^nC_6 \left(3^{\frac{1}{3}}\right)^{n-6} \cdot \left(4^{\frac{-1}{3}}\right)^6 \\
 Q &= {}^nC_{n-6} \left(3^{\frac{1}{3}}\right)^6 \cdot \left(4^{\frac{-1}{3}}\right)^{n-6} \\
 \therefore \frac{Q}{P} &= 12 \Rightarrow (12)^{\frac{n-6}{3}} = (12)^1 \\
 \Rightarrow \frac{n-6}{3} &= 1 \Rightarrow n = 9
 \end{aligned}$$

9. Ans (2.00)

$$\begin{aligned}
 k &= \lim_{x \rightarrow 0} \frac{\sqrt{1 + \sin^2(x^2)} - \cos^3(x^2)}{x^4} \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{1 + \sin^2(x^2) - \cos^6(x^2)}{x^4} \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{(1 - \cos^3(x^2))(1 + \cos^3(x^2))}{x^4} + \underbrace{\frac{\sin^2(x^2)}{x^4}}_{=1} \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \left[\frac{(1 - \cos x^2)(1 + \cos x^2 + \cos^2(x^2))(1 + \cos^3(x^2))}{x^4} + 1 \right] \\
 &= \frac{1}{2} \left[\frac{1}{2} \times 3 \times 2 + 1 \right] = 2 \text{ Ans.}
 \end{aligned}$$

10. Ans (72.00)

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x+0)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x) + f(h) + h\sqrt{f(x)} - \{f(x) + f(0) + 0\sqrt{f(x)}\}}{h} \\
 &= \lim_{h \rightarrow 0} \left\{ \frac{f(h) - f(0)}{h} \right\} + \sqrt{f(x)} \\
 &= f'(0) + \sqrt{f(x)} \\
 &= 0 + \sqrt{f(x)} = \sqrt{f(x)} \\
 \text{So,} \\
 \int \frac{f'(x)}{\sqrt{f(x)}} dx &= \int dx \Rightarrow 2\sqrt{f(x)} = x + c \\
 (c = 0, \text{ as } f(0) = 0) \\
 \Rightarrow f(x) &= \frac{x^2}{4}
 \end{aligned}$$

Note that when $\alpha = 0$ area is minimum.

$$\text{Required minimum area} = 2 \int_0^9 2\sqrt{y} dy = 72$$